



MILLET COOKER:

An Electric Slow Cooker

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In 2023, we celebrate the International Year of Millets, honouring the nutritional richness and versatility of millets, such as Foxtail, Barnyard, Kodo, Little, and Brown-top varieties. These grains are known for their high dietary fibre content and diverse nutrient profiles. To harness their benefits fully, it is vital to ensure minimal nutrient loss during cooking while achieving the desired softness for easy digestion.

Soak millets for at least four hours before cooking, focusing on one variety at a time to prevent overflow when using a pressure cooker. Our solution: design of a slow cooker that cooks nutritious, soft, and tasty millets perfectly. This cooker is energy-efficient, keeping food warm due to thermal insulation, with cooking times ranging from 1 to 1.5 hours.

The block diagram of the proposed cooker is shown in Fig. 1. It

employs an 1800W electric heater coil powered by a 48V DC output from an AC-DC converter. An N-channel MOSFET controls the power supply to the heater. The heater calculations are as follows:

1. Resistance of heater coil = $(230 \times 230)/1800 = 29.4\Omega$
2. Heat generated by coil = $(48 \times 48)/29.4 = 78.4W$
3. Current through the coil = $48/29.4 = 1.6A$

Therefore, AC-DC converter having current rating of 2.3A is selected.

Two parameters determine the cooking time. The first is the heater temperature, which is monitored using a thermistor connected to a comparator circuit. The comparator trip point is set using a potmeter (VR2). This is a onetime setting. When the thermistor reaches the set temperature, the comparator enables the counter through RST signal. The second parameter is the user-controlled counter time setting. It works on setting the clock frequency to the input of the counter. Once the counter overflows, the MOSFET is turned off. The sum of both timings determines the total cooking time.

Having two parameters to determine cooking time offers flexibility. The temperature sensing accounts for the quantity of millets, allowing quicker heating for smaller quantities and longer times for larger ones. The users

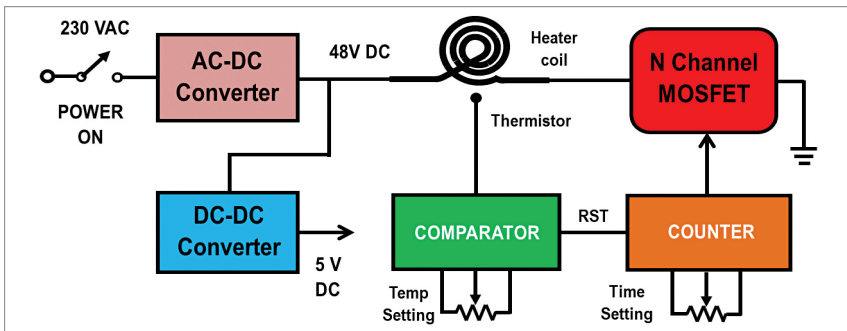


Fig. 1: Block diagram of electric millet slow cooker

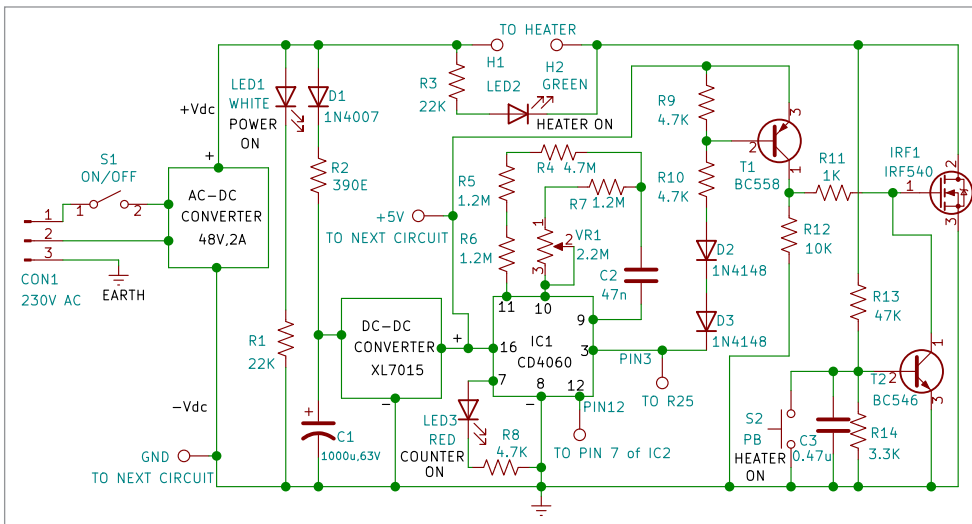


Fig. 2: Circuit diagram of the power supply, counter, and MOSFET sections